Major Optical Transport Failure Resolution for ELEC-PE-06 at Electronic City Hub using Ericsson CLI

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_5287

Description

The failure domain of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the PE role, specifically on Ericsson equipment, occurs due to the inherent vulnerability of optical transceivers to physical damage or degradation. The Optical Transport Network (OTN) protocol, which operates on the data plane, relies on the integrity of the optical signals transmitted between nodes. Any degradation in the optical signal quality, caused by SFP/XFP malfunction or fiber cuts, can lead to signal loss and network instability. Furthermore, the Remote Procedure Call (RPC) mechanism, used for control plane communication, may experience latency or packet loss due to the affected optical signals. Consequently, the PE node's ability to establish and maintain connections with neighboring nodes is compromised, resulting in the ALARM_5287 alarm code being triggered. This standard operating procedure outlines the steps necessary to diagnose and resolve such a failure, ensuring minimal disruption to the network.

Pre-requisite Checks

- Check: Verify the site location and ensure the Electronic City Hub is physically accessible for remote hands dispatch.

- Check: Confirm the redundant link status and ensure it is functioning correctly.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to manufacturing defects or aging.	High
Fiber cuts or physical damage caused by human error or environmental factors.	Medium
Rx/Tx power loss due to optical signal attenuation or interference.	Low

Diagnostic Commands & Expected Outputs

```
Command: admin show optical-transceiver status, Expected Output: Optical Transceiver Status: OK, Description: This command verifies the operational status of the optical transceiver, confirming whether it is functioning correctly or not.

Command: admin show otu-control-plane status, Expected Output: OTU Control Plane Status: OK, Description: This command checks the status of the OTU control plane, ensuring it is operational and communicating correctly with neighboring nodes.
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```
Command: admin show otu-data-plane status, Expected Output: OTU Data Plane Status: OK, Description: This command verifies the operational status of the OTU data plane, confirming whether it is functioning correctly or not.
```

Resolution Steps

Step 1: Step: Swap SFP/XFP with a spare unit., Command: admin replace optical-transceiver , Description: This step involves replacing the faulty SFP/XFP with a spare unit to restore optical signal integrity and network stability.

Step 2: Step: Inspect and repair fiber cuts or physical damage., Command: admin repair fiber-cut , Description: This step involves inspecting the affected fiber and repairing any cuts or physical damage to restore optical signal quality and network stability.

Step 3: Step: Adjust Rx/Tx power levels to optimal values., Command: admin adjust rx-tx-power , Description: This step involves adjusting the Rx/Tx power levels to optimal values to minimize signal attenuation and interference.

Post-Check Verification

```
Command: admin show optical-transceiver status, Expected Output: Optical Transceiver Status: OK, Description: This command verifies the operational status of the optical transceiver, confirming whether it is functioning correctly or not.

Command: admin show otu-control-plane status, Expected Output: OTU Control Plane Status: OK, Description: This command checks the status of the OTU control plane, ensuring it is operational and communicating correctly with neighboring nodes.

Command: admin show otu-data-plane status, Expected Output: OTU Data Plane Status: OK, Description: This command verifies the operational status of the OTU data plane, confirming whether it is functioning correctly or not.
```

Roll-back Procedures

- Step: Revert SFP/XFP swap., Command: admin revert optical-transceiver , Description: This step involves reverting the SFP/XFP swap to its original state if the fix fails.

- Step: Re-inspect and re-repair fiber cuts or physical damage., Command: admin revert repair fiber-cut , Description: This step involves reverting the fiber repair to its original state if the fix fails.

- Step: Re-adjust Rx/Tx power levels to original values., Command: admin revert adjust rx-tx-power , Description: This step involves reverting the Rx/Tx power level adjustment to its original state if the fix fails.